Spatial Heterogeneity of Drought Shocks and Irrigation Buffers in U.S. Alfalfa Trade

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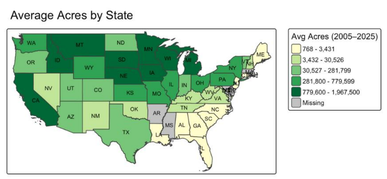
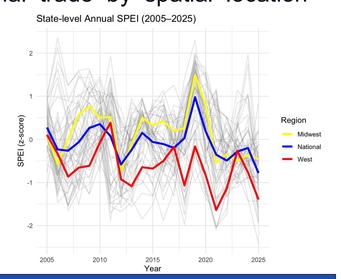
Introduction:

Understanding Alfalfa:
Alfalfa is the third most valuable field crop nationally (about 23 million acres harvested, $9 billion annually) and is paradoxically both drought tolerant biologically and water intensive agronomically. Drought pressure increasing in the United States, especially alfalfa intensive region (West and Mid-West).

Purpose:
This paper integrates spatial drought diagnostics with econometric modeling to quantify how drought affects alfalfa yields and exports, and to assess whether irrigation and international trade by spatial location—buffer these impacts.





Background:

Drought threatens the stability of the United States' forage system, yet the magnitude and spatial heterogeneity of its impacts on alfalfa have not been illustrated. And alfalfa is central to water debates in the West due to higher drought pressure and intensive water demand for export market.

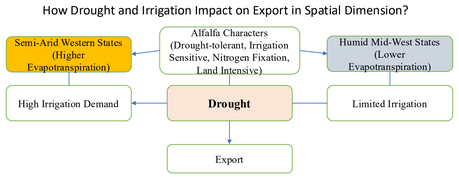
Commodity	% Area Affected
Alfalfa Hay Acreage	32
Hay Inventory	33
Milk Cow Inventory	39
Cattle Inventory	25

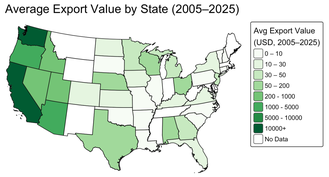
Source: NOAA-NIDIS, updated in Oct. 2025.

Research Questions:

Environment/Water policy scientist arguments:
Cantor et al., (2022) argue that alfalfa delivers comparatively low economic returns per unit of water ("dollars per drop" and "jobs per drop"). When this hay is exported, the region is essentially exporting its scarce water (Fu et al., 2023).

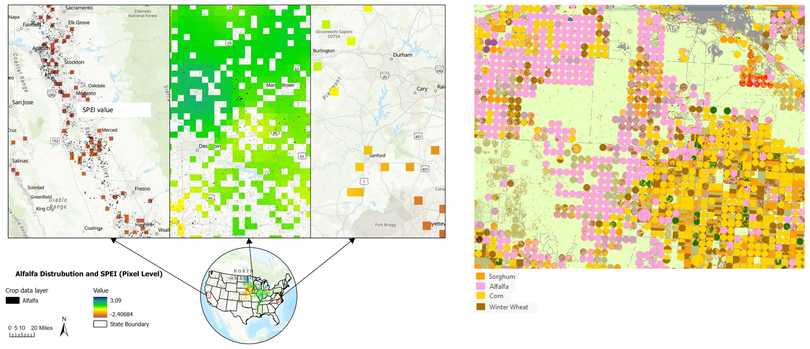
Agronomist arguments:
Alfalfa can enhance agricultural system resiliency because it is relatively "flexible": producers can readily idle (fallow) stands or reduce irrigation when water becomes scarce, making alfalfa a manageable margin of adjustment during drought (Yost et al., 2020; Fleck, 2016). Alfalfa also provides ecosystem services (nitrogen concentration) and is necessary for land rotation in the core belt.





Methodology:

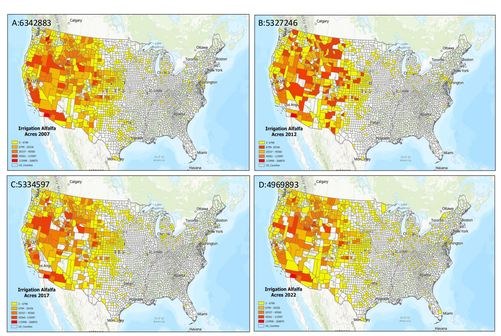
- Uused state-year panel that links the SPEI, alfalfa yield, export, irrigation and integrates geospatial diagnostics with econometric models.
- SPEI-03 (averaged Apr—Sep), which aligns with the alfalfa's growing and cutting season, captures short to seasonal moisture deficits that directly affect forage regrowth and yield formation, and incorporates atmospheric water demand through evapotranspiration rather than rainfall alone.
- Combined pixel-level SPEI data with USDA Cropland Data Layer (CDL) information to identify drought conditions specifically on alfalfa land. It can reduce aggregation bias and and allow the drought measure to better reflect within-state variation in the drought conditions experienced by alfalfa acreage

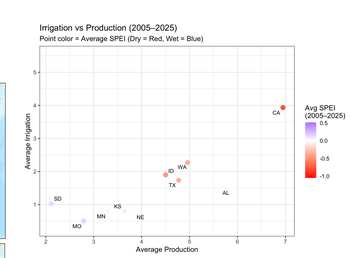


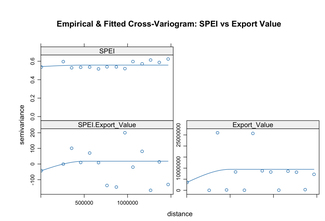
Spatial Panter:

Irrigated alfalfa is structurally a Western crop:
Spatial "hotspots" are persistent over time (path dependence). The same production corridors remain prominent across panels—consistent with irrigation being tied to fixed infrastructure, water rights, and agro-climatic comparative advantage, rather than short-run switching.

Export-SPEI spatial decoupled:
Western states cluster under negative SPEI while maintaining high yields. This spatial divergence highlights how irrigation can mediate drought impacts.







Results:

- Export value does not respond systematically to drought shocks in reduced form once fixed effects absorb persistent heterogeneity.
- 2SLS evidence supports a production-capacity channel: instrumented yield significantly increases export value, indicating drought influences exports primarily through yield impacts rather than directly.
- Export patterns are anchored more in structural location advantages than in contemporaneous moisture conditions. Drought is a material and spatially clustered production risk for U.S. alfalfa.

	Cluster(state)	One-way FE	Two-way FE	One-way FE	WLS
Dependent variable	Yield/acre	Yield/acre	Yield/acre	log(production)	log(production)
SPEI	−0.0071	−0.0069†	0.1144***	0.0436	0.4525*
Irrigation	0.6202***	1.1945***	0.1636†	0.3413*	0.8631**
West	1.444***	1.0056***		3.5098***	3.0375***
Midwest	0.0287***	0.2438***		3.5660***	2.5961***
Fixed effects	No	Year	State + Year	Year	No
Observations	903	903	903	903	149

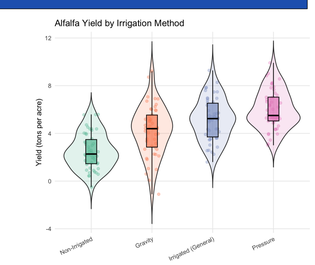
Notes: $^{\dagger}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. SPEI remains positive and significant across robustness checks.

	(1) One-way FE	(2) OLS	(3) GAM (Spatial)	(4) 2SLS
Dependent variable	Export Value	Export Value	Export Value	Export Value
SPEI	269.7	303.5	329.8	−84.47†
Irrigation	267.5	−408.7	253.7†	1836***
Alfalfa yield	804.0***	919.0***	1189***	1997***
West	2102**	3217***	—	2242.6***
Price effect	20.40***	16.24***	s(Price) sig. ($p = 0.028$)	20.92**
Spatial smooth	No	No	s(lon,lat) sig. ($p < 2 \times 10^{-16}$)	No
Fixed effects	Year	No	No	State + Year
Model fit	Adj. $R^2 = 0.309$	Adj. $R^2 = 0.296$	Adj. $R^2 = 0.808$ (81.5% dev.)	Adj. $R^2 = 0.280$
Observations	619	619	619	619

Notes: GAM reveals nonlinear price effects and strong spatial structure.

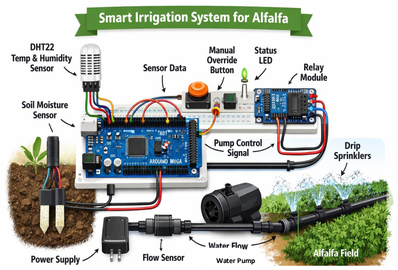
Policy Implication:

Sales contract and water rights allocation practices left many growers with little incentive to change growing practices. Pressure irrigation exhibits the lowest variability and the highest ton-per-acre-foot efficiency, underscoring irrigation's role as a key drought adaptation—while also highlighting exposure to tightening water constraints.



Treatment	Mean Yield	SD	CV (%)	Mean-Eff	Yield Gain (%)
Non-Irrigated	2.47	1.52	66.1	0	0
Gravity Irrigation	4.31	2.02	47.1	3.36	74.5
Irrigated (Combined)	4.53	2.23	48.6	8.78	82.6
Pressure Irrigation	4.58	1.72	38.5	10.13	85.4

Notes: Mean yield is measured in tons per acre. CV denotes the coefficient of variation. Group letters indicate multiple-comparison results; treatments sharing no common letter differ at the 5% significance level. Mean_Eff is average efficiency (tons per acre-foot). Percentage gains are calculated relative to the non-irrigated baseline.

Conclusion:

The stability/efficiency advantage of pressure drip systems suggests gains from smart water management, while maintaining alfalfa yield and planning should explicitly consider water policy and groundwater constraints.